

Predictive Models for Educational Purposes: A Systematic Review

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Abstract: This systematic literature review evaluates predictive models in education, focusing on their role in forecasting student performance, identifying at-risk students, and personalising learning experiences. The review compares the effectiveness of machine learning (ML) algorithms such as Support Vector Machines (SVMs), Artificial Neural Networks (ANNs), and Decision Trees with traditional statistical models, assessing their ability to manage complex educational data and improve decision-making. The search, conducted across databases including ScienceDirect, IEEE Xplore, ACM Digital Library, and Google Scholar, yielded 400 records. After screening and removing duplicates, 124 studies were included in the final review. The findings show that ML algorithms consistently outperform traditional models due to their capacity to handle large, non-linear datasets and continuously enhance predictive accuracy as new patterns emerge. These models effectively incorporate socio-economic, demographic, and academic data, making them valuable tools for improving student retention and performance. However, the review also identifies key challenges, including the risk of perpetuating biases present in historical data, issues of transparency, and the complexity of interpreting AI-driven decisions. In addition, reliance on varying data processing methods across studies reduces the generalisability of current models. Future research should focus on developing more transparent, interpretable, and equitable models while standardising data collection and incorporating non-traditional variables, such as cognitive and motivational factors. Ensuring transparency and ethical standards in handling student data is essential for fostering trust in AI-driven models.



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Keywords: big data; machine learning; algorithms; artificial intelligence; predictive models; statistical models; education; systematic literature review; SLR

1. Introduction

1.1. Rationale and Background

In recent years, there has been a significant advancement in the utilisation of artificial intelligence (AI) in various fields, including education. This technology is not just reshaping industries; it is also helping educators address some of the biggest challenges in the classroom. This progress has been primarily driven by the development of deep neural network (DNN)-based AI technologies, which have facilitated the incorporation of robust computational capabilities, state-of-the-art algorithms, and extensive data processing capacities [1]. Among the subfields of AI, ML has emerged as a key area, offering a variety of algorithms that deliver substantial advantages. ML methods such as Support Vector Machines and Naive Bayes (NB) classifiers demonstrate the practical value of ML in solving complex challenges [2].

Machine learning models play a critical role in predictive analytics, particularly in projecting outcomes across diverse domains, such as science, engineering, and healthcare, aiming to forecast future events and trends based on past data analysis [3–7]. Machine

learning-powered predictive models have moved beyond just theoretical concepts; they are now practical tools that directly align with core educational objectives. Two areas where they have made a significant impact are helping students stay engaged and on track in terms of retention and creating learning experiences tailored to individual needs through personalised learning [8,9].

Student retention is a crucial challenge that predictive models address effectively. Every year, many students disengage or drop out, often because early signs of struggle go unnoticed. By analysing patterns in data such as attendance, grades, and behavioural trends, predictive models can flag at-risk students early. This allows schools to step in with targeted support, such as mentoring or tutoring, before it is too late. Research shows that these early interventions can significantly boost retention rates and academic success [10,11].

Another way predictive models are changing education is by enabling personalised learning. Traditional teaching methods often fail to meet the diverse needs of students, leading to inconsistent outcomes and reduced engagement. ML algorithms, such as Decision Trees and neural networks, help educators understand how individual students learn best. For example, some students might benefit from extra visual aids, while others excel with hands-on projects. By tailoring teaching strategies to these unique needs, predictive models make learning more effective and enjoyable [9,12].

These models also enable institutions to allocate their resources more effectively. For instance, predictive analytics can identify which students might benefit the most from additional counselling or academic support, ensuring that resources are used where they are needed most. In online learning environments, these tools can highlight challenging course content or spot trends in student engagement, helping instructors refine their approaches dynamically [10].

Moreover, predictive models promote equity and inclusivity in education. By identifying barriers such as socio-economic or demographic factors that might affect student success, these tools enable educators to design targeted interventions that close achievement gaps. For example, they can highlight disparities in resource access and guide schools in developing strategies to support underrepresented groups [9].

The development and success of predictive models rely on advanced methods such as feature extraction algorithms and attention-based networks, which integrate multiple machine learning techniques for forecasting [13,14]. Deep learning algorithms, in particular, excel in analysing complex educational data. Their ability to identify patterns and learn from examples has allowed these models to outperform traditional methods, significantly enhancing prediction accuracy [11]. Innovative approaches, such as analysing procrastination behaviours, have further expanded the potential of predictive models, providing deeper insights into student performance and the factors influencing academic success [15].

1.2. Objectives

This paper presents a systematic literature review on Predictive Models for Educational Purposes through evaluation and comparison of these models. The paper contributes by providing answers to the following seven questions:

- What are the key reasons for using prediction models in education, and how do they impact student outcomes?
- What are the most used datasets for predicting student performance in educational predictive models?
- What are key factors influencing the accuracy of predictive models in an educational context?
- How do prediction models in machine learning enable accurate forecasting and decision-making in education?
- How do prediction models account for uncertainty and variability in their forecasts in education?

- How do machine learning algorithms compare to traditional statistical models in predicting student outcomes?
- What are the potential applications and implications of using prediction models in education, and what are the possible risks and challenges?

1.3. Contributions

The following are the major contributions of this paper:

- This paper provides a systematic literature review of predictive models in education, focusing on their application in forecasting student performance, identifying at-risk students, and personalising learning experiences. This offers a detailed overview of the state of the art in the field.
- The study evaluates and compares the effectiveness of various ML algorithms, such as SVMs, ANNs, and Decision Trees, against traditional statistical models. It highlights the superior capacity of ML models to handle large, non-linear datasets and improve predictive accuracy.
- Key challenges in the adoption of predictive models are identified, including biases in historical data, issues of transparency and interpretability, and the lack of standardisation in data preprocessing methods.
- The paper suggests directions for future research, such as developing interpretable and equitable models, standardising data collection processes, and incorporating non-traditional variables, such as cognitive and motivational factors, into predictive frameworks.
- The review discusses practical implications for the use of predictive models in education, such as improving resource allocation, designing targeted interventions, and promoting equity through data-driven strategies.
- It emphasises the importance of ethical standards and transparency in the implementation of AI-driven predictive models to foster trust among stakeholders in education.

2. Methodology

This systematic review follows the PRISMA guidelines and adheres to the methodologies proposed by Kitchenham [16] and illustrated in Supplementary Materials.

2.1. Information Sources and Search Strategy

As part of this literary study, publications from well-attended conferences, journals, and books, covering the period from 1 January 2000 to 1 July 2024 were examined. The time frame was selected to capture advancements in predictive modelling and machine learning in education, coinciding with significant technological progress in data analytics, artificial intelligence, and educational technology during this period. The year 2000 marks the beginning of widespread adoption of learning management systems (LMSs) and data collection in education, providing a robust dataset for evaluating predictive models.

To identify relevant literature, four high-quality academic databases were used: ScienceDirect, IEEE Xplore, ACM Digital Library, and Google Scholar via ACM. These databases were selected for their comprehensive coverage of peer-reviewed research in computer science, artificial intelligence, and educational technology. ScienceDirect provides multidisciplinary insights into educational technology and AI applications; IEEE Xplore focuses on engineering and machine learning research; ACM Digital Library provides extensive studies in computer science and educational data mining; and Google Scholar via ACM ensures access to additional relevant studies not covered by traditional databases.

A comprehensive search was conducted using defined keywords related to the focal topic. These terms included the significance of forecasting models in the sector of education, educational predictive modelling, predictive analytics in education, machine learning for student performance, factors impacting the precision of educational prediction models, precise forecasting, decision-making in educational prediction models, uncertainty and variability in educational prediction models, conventional statistical models in edu-

cation, and applications, implications, risks, and challenges associated with educational prediction models.

2.2. Study Selection

The initial search yielded 400 publications, comprising books, conference papers, book chapters, and peer-reviewed journal articles. A two-step process was then implemented to screen the articles. The first step involved evaluating titles, abstracts, keywords, introductions, and conclusions to assess their relevance to the research topic, resulting in the exclusion of articles based on duplication, language limitations, repeated publishing across different databases, and divergence from the subject matter.

2.3. Inclusion and Exclusion Criteria

During the final stage of the selection process, three additional criteria were applied to ensure that studies met the specific requirements for predictive models in educational data:

1. **Alignment with the Research Theme:** Articles had to address at least one of the core research questions, ensuring their relevance to predictive models in education and their specific applications.
2. **Educational Focus:** The paper prioritised studies that specifically explored applications in the educational field, as this focus aligns with the unique challenges and goals of educational data prediction such as improving student retention and personalising learning.
3. **Robust Methodology:** Only studies employing rigorous research methods—whether qualitative, quantitative, case studies, or focus groups—were included. This ensured the findings were reliable and could contribute meaningfully to understanding predictive model applications in education.

The complete process is pictured in Figure 1 and the results are discussed in Section 3.1. The number of articles addressing each of the research questions outlined in Section 1.2 is summarised in Appendix A Table A1, while the distribution of the included studies is illustrated in Appendix A Table A2.

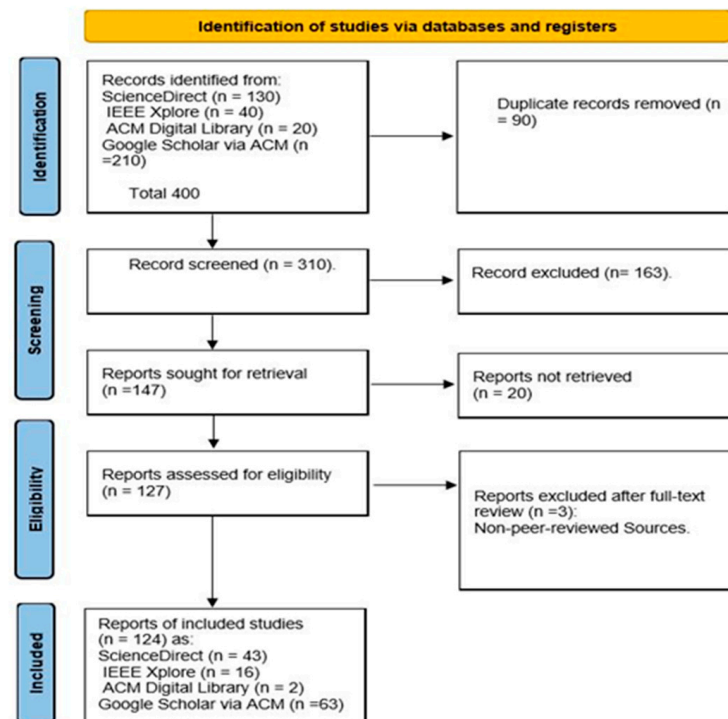


Figure 1. Article selection and analysis flow.

3. Results

3.1. Study Selection and Characteristics

The search across multiple databases yielded a total of 400 records, sourced from ScienceDirect ($n = 130$), IEEE Xplore ($n = 40$), ACM Digital Library ($n = 20$), and Google Scholar via ACM ($n = 210$). After removing 90 duplicates, 310 unique records remained for further screening. At the screening phase, 163 records were excluded based on title and abstract review, leaving 147 full-text articles for eligibility assessment.

Among these, 20 reports could not be retrieved, and 127 articles were thoroughly assessed for eligibility based on inclusion criteria such as relevance to machine learning-based height prediction, sample size, and study design. Ultimately, 3 articles were excluded as they were not peer-reviewed, resulting in a final sample of 124 studies.

3.2. Risk of Bias

The risk of bias assessment revealed several key issues that affected the results and reliability of the predictive models. Selection bias was prevalent, with many studies relying on datasets that were not representative of the broader population, such as focusing exclusively on high-performing students or specific demographic groups. This limitation limited the generalisability of the results and often led to overly optimistic predictions. Reporting bias was also a concern, with some studies lacking transparency in describing data collection methods, handling of missing data, and variable selection processes. These omissions created uncertainty regarding the quality and reliability of the results.

Publication bias also affected the results, with studies reporting significant results more likely to be published, leading to an overestimation of the effectiveness of the model. Algorithmic bias was another notable issue, with models trained on data that reflected socio-economic or demographic inequalities sometimes perpetuating these inequalities, raising ethical concerns and reducing fairness.

Despite these challenges, many studies demonstrated methodological rigour through clear documentation of data sources, transparent preprocessing, and robust validation techniques. The exclusion of non-peer-reviewed studies contributed to the reliability of the review by ensuring adherence to established academic standards.

3.3. Results of Individual Studies and Synthesis

This systematic review aimed to address the seven key research questions (outlined in Section 1.2) surrounding the use of prediction models in education, focusing on their application, performance, and implications.

1. What are the key reasons for using prediction models in education, and how do they impact student outcomes?

Prediction models play a crucial role in education by enhancing the ability to forecast student performance, particularly identifying at-risk students early for timely intervention. Studies show that integrating socio-economic, demographic, and behavioural data improves the ability to predict student outcomes and inform personalised learning strategies. The models enable educators to allocate resources more effectively, identify students requiring additional support, and ultimately improve academic achievement [9,17].

2. What are the most used datasets for predicting student performance in educational predictive models?

This review identified that datasets commonly used for predicting student performance include a combination of socio-economic, demographic, behavioural, and academic metrics. E-learning analytics data, such as homework completion rates, participation in LMS, and attendance, were widely used across studies. In addition, publicly available datasets from repositories such as the UC Irvine Machine Learning Repository were frequently used in model development, alongside institution-specific datasets [9,18].

3. What are the key factors influencing the accuracy of predictive models in an educational context?

Several factors were identified as critical in influencing the accuracy of predictive models in education. The choice of machine learning algorithms, such as Gradient Boosting, Random Forest, and ANNs, was found to be particularly important for handling non-linear relationships and high-dimensional data [19]. Data quality, preprocessing methods (e.g., normalisation and imputation), feature selection techniques (e.g., Principal Component Analysis), and model validation methods (e.g., cross-validation) were also significant determinants of accuracy [20,21].

4. How do prediction models in machine learning enable accurate forecasting and decision-making in education?

ML algorithms, such as SVMs and ANNs, demonstrated superior ability in managing large, complex datasets and capturing intricate, non-linear patterns in student behaviour. These capabilities enable more precise forecasting of student outcomes and support data-driven decision-making in education [22]. Additionally, automated machine learning (AutoML) systems streamline model selection and hyperparameter tuning, enhancing the overall accuracy of predictions and supporting proactive student interventions [10].

5. How do prediction models account for uncertainty and variability in their forecasts in education?

This review found that prediction models address uncertainty and variability through techniques such as Bayesian probabilistic forecasting and Empirical Best Linear Prediction (EBLP), which account for variability across different datasets and reduce mean squared error. Methods such as Monte Carlo simulations and probabilistic forecasting were also effective in providing confidence intervals, allowing educators to better understand the range of potential outcomes and improve decision-making under uncertainty [23,24].

6. How do machine learning algorithms compare to traditional statistical models in predicting student outcomes?

Machine learning algorithms consistently outperformed traditional statistical models, such as linear regression, in predicting student outcomes. ML models were more flexible in handling large, complex datasets and adapting to new data patterns. Deep learning models such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks were particularly effective in identifying complex patterns in student behaviour, contributing to more reliable predictions in various educational settings, including MOOCs and hybrid learning environments [25].

7. What are the potential applications and implications of using prediction models in education, and what are the possible risks and challenges?

While prediction models offer numerous applications in enhancing student outcomes and optimising institutional decision-making, several challenges were identified. These include the potential for bias in models trained on historical data, which may perpetuate inequalities related to socio-economic status, race, and gender. Privacy concerns related to the extensive use of personal data, the complexity of “black box” AI models, and risks of misclassification were also noted as key issues requiring ongoing scrutiny [26,27]. Additionally, ensuring ethical use of these models is essential to maintain trust and promote fairness in education.

3.4. Data Type Reliability in Predictive Modelling

Among the three data types—behavioural, academic, and socio-economic—behavioural data consistently emerged as the most reliable across studies. Metrics such as class attendance, participation in LMS, and study habits provided real-time, actionable insights into student engagement and learning behaviours. These indicators were particularly effective

for identifying at-risk students and accurately predicting short-term outcomes, such as course completion rates [9,18].

While academic data, such as grades and test scores, provided a solid foundation for long-term performance predictions, their reliability often increased when combined with behavioural data to capture a more comprehensive view of student performance [9,20].

Meanwhile, socio-economic data, while critical for understanding broader external factors and systemic barriers, were less directly tied to immediate performance metrics. Their meaningful integration required careful contextual analysis to avoid introducing biases. Nevertheless, socio-economic data played a vital role in promoting equity and designing targeted interventions, ensuring that models addressed disparities in learning opportunities and outcomes [17,18].

3.5. Subgroup Analyses and Sensitivity Analyses

Subgroup analyses were conducted to explore variations in the performance of predictive models based on different types of data and machine learning techniques. For example, studies using behavioural data alongside traditional academic performance metrics demonstrated more accurate predictions of student success. In addition, studies focusing on online learning environments reported higher model efficacy due to the granularity of data from LMS.

Sensitivity analyses were performed by excluding studies with higher risks of bias to assess the robustness of the findings. The exclusion of these studies did not significantly alter the overall results, indicating that the conclusions of the review are stable and reliable. Moreover, models that employed cross-validation techniques were found to be more consistent in predicting student performance.

3.6. Certainty of Evidence

The certainty of the evidence was evaluated based on the study designs and methodologies employed across the included studies. While many studies provided robust data and methodologies, some were downgraded due to incomplete reporting, small sample sizes, or potential biases in study design. Despite these limitations, the body of evidence indicates that predictive models in education are effective in improving student outcomes, particularly in settings such as online learning environments.

3.7. Comparing Model Performance Across Educational Settings

The study characteristics in Appendix A Table A3 reveal notable variations in the efficacy and suitability of predictive models across different educational contexts such as K-12, higher education, and online learning environments [28–53]. The following highlight how the choice of predictive models must align with the specific challenges and opportunities of the educational context to maximise efficacy and ethical implementation.

K-12 settings: Neural networks, Bayesian models, and machine learning techniques are prominently applied in K-12 contexts. Studies such as [22,54] highlight the robustness of machine learning models in predicting attendance and academic performance. Bayesian models, such as [23], effectively forecast long-term trends in academic outcomes, supporting data-driven planning. Ethical challenges in AI integration, as noted by [26], are particularly relevant in K-12 settings where fairness and transparency are critical concerns. Data limitations in smaller K-12 datasets may reduce model accuracy compared to larger-scale studies in higher education.

Higher education settings: Gradient Boosting, Random Forests, deep learning, and hybrid approaches dominate in higher education contexts. Studies such as [9] demonstrate the advantages of explainable ML models for enhancing decision-making transparency. Ensemble models and hybrid approaches (e.g., [31,55]) achieve high predictive accuracy, supporting proactive interventions for retention and performance. Data-driven approaches, such as [56], leverage diverse datasets to provide actionable insights into student success predictors. The trade-offs between interpretability and accuracy, as discussed in [55],

remain a challenge. Issues with imbalanced datasets and socio-economic variability, as noted by [57], can limit model generalisability.

Online and MOOC settings: Gradient Boosting and deep learning are widely used due to their scalability and precision. Gradient Boosting models (e.g., [58,59]) excel in identifying predictors of retention and engagement, which are critical in online environments. The integration of big data and interaction analytics, such as [27], enables scalable predictions in MOOCs and e-learning platforms. The reliance on engagement and interaction data can lead to overfitting if not properly validated across diverse populations. Ethical concerns related to student data privacy and algorithm transparency, particularly in large-scale MOOCs, need careful attention.

Insights and implications: K-12 settings benefit from simpler and interpretable models, while higher education and MOOCs require advanced algorithms to handle the complexity of larger datasets. Tailored strategies are essential. For K-12, ethical AI guidelines and fairness considerations should accompany model deployment. In higher education and MOOCs, leveraging ensemble and hybrid models can provide robust insights but must be paired with transparent frameworks to build trust. Future studies should explore cross-context comparisons to identify transferable model features and develop frameworks that adapt to the unique needs of each educational setting.

4. Discussion

This section provides a deeper analysis of the results and findings presented in Section 3.3, examining them in relation to the seven research questions outlined in Section 1.2. By revisiting each of the research questions, this paper aims to critically assess how the data and outcomes address the key objectives of this study. This discussion will highlight any patterns, correlations, or unexpected results that emerged, offering insights into both the strengths and limitations of the findings. In addition, the research will explore how these results align with the existing literature, where they diverge, and what implications they hold for future research or practical applications. The aim is to provide a comprehensive interpretation of the data in relation to the research questions, shedding light on the broader significance of this study.

4.1. Key Reasons for Using Prediction Models in Education and Their Impact on Student Outcomes

This review identifies several key roles that prediction models play in education, particularly in enhancing the academic outcomes of students. The primary aim of these models is to forecast student performance, allowing educators to identify at-risk students and implement early interventions that can significantly boost their chances of academic success [17]. In line with previous studies, this review highlights that prediction models are particularly effective when they incorporate socio-economic, demographic, and psychological factors, which enables the creation of tailored intervention strategies [60]. Such strategies can not only improve student outcomes but also contribute to the refinement of resource allocation and instructional methods, aligning with findings from Kumar et al. [61]. This review underscores the value of these models in specific educational settings, such as remedial education, where early interventions based on data-driven insights have the potential to dramatically improve student trajectories [62].

Moreover, the integration of artificial intelligence and machine learning techniques in prediction models optimises educational processes, particularly in online course design, by identifying the most impactful factors contributing to student success [63]. While these models are robust in improving instructional design and fostering personalised learning experiences, there remains a need for further empirical evidence on their long-term impact on educational policy and practices.

This systematic review further emphasises the significant impact prediction models have on student outcomes. These models enhance the ability of educational institutions to detect at-risk students early, allowing for timely interventions that improve academic performance and retention rates [9,64]. Consistent with the literature, this review finds that

models using socio-economic and pre-course performance data, among other variables, can effectively support personalised learning approaches [9,65]. Furthermore, the integration of machine learning into prediction models has been shown to enhance the precision of early alerts, making it possible to implement targeted interventions at the beginning of academic terms, which may help prevent student dropouts [64,66].

However, while the models have demonstrated predictive power, the degree of their accuracy and effectiveness varies depending on the quality and comprehensiveness of the input data. Some studies suggest that traditional factors, such as high school GPA and standardised test scores, may not be universally reliable predictors, particularly in diverse educational settings [67,68]. Thus, future research should focus on validating the predictive power of newer data inputs, such as cognitive and motivational factors, across varied educational contexts.

Data-driven insights generated by educational prediction models have proven to be valuable in improving institutional decision-making. Through advanced machine learning and data mining techniques, these models provide deep insights by processing large datasets that include socio-demographic variables, academic records, and student engagement metrics. These insights enable institutions to pinpoint critical factors affecting student success, such as early academic performance, which studies have shown to be a strong predictor of future achievement [69]. By utilising this information, institutions are better positioned to take proactive measures to support students on the edge of academic risk, thereby improving retention rates and graduation outcomes [70].

Furthermore, the review finds that the use of artificial intelligence in prediction models enhances institutional resource allocation by improving the precision of performance forecasts [71]. However, while the models offer significant improvements over traditional predictors, there is a need for more comprehensive and inclusive approaches that consider a wider range of cognitive, motivational, and behavioural factors [56]. This expanded approach could help educational institutions better understand student needs and respond with personalised, data-informed strategies that enhance both student performance and institutional policies.

One of the most important contributions of prediction models is their ability to enable proactive student support. This systematic review highlights that by identifying at-risk students early, institutions can implement timely interventions that are customised to meet the specific needs of students, thereby improving retention rates and academic outcomes [72,73]. By classifying students into different risk levels, educators can direct their efforts toward those who require the most assistance, ensuring that vulnerable populations receive the necessary support [74].

In addition, the review finds that prediction models incorporating non-cognitive factors, such as perseverance and social support networks, offer a more holistic understanding of the factors that influence student success [75]. This aligns with recent studies that suggest that non-cognitive variables can play a crucial role in predicting student outcomes, particularly in diverse educational contexts.

4.2. Most Used Datasets for Predicting Student Performance in Educational Models

This systematic review has identified the most commonly used datasets for predicting student performance in educational predictive models. The findings highlight several recurring types of data and the importance of socio-economic, behavioural, and academic metrics in these models.

A significant proportion of studies used datasets that combine socio-economic factors, such as family background and parental education, with academic performance data [9]. E-learning analytics data, including metrics from LMS such as homework completion and class attendance, also play a pivotal role in predictive models [9]. Notably, many models leverage a combination of numerical and categorical data to capture diverse aspects of student behaviour, though some rely on a single data type.

Datasets sourced from repositories, such as the UC Irvine Machine Learning Repository, which incorporate demographic, social, and academic features, were among the most commonly used [18]. In addition, university-specific datasets, such as those from George Mason University [76] and Can Tho University in Vietnam [77], have been instrumental in building predictive models that focus on student performance. These datasets vary in terms of size, structure, and the type of attributes included, allowing for flexibility in model development depending on the educational context.

The datasets illustrated in Figure 2 represent the range of data types and sources used to predict student outcomes in different educational contexts.

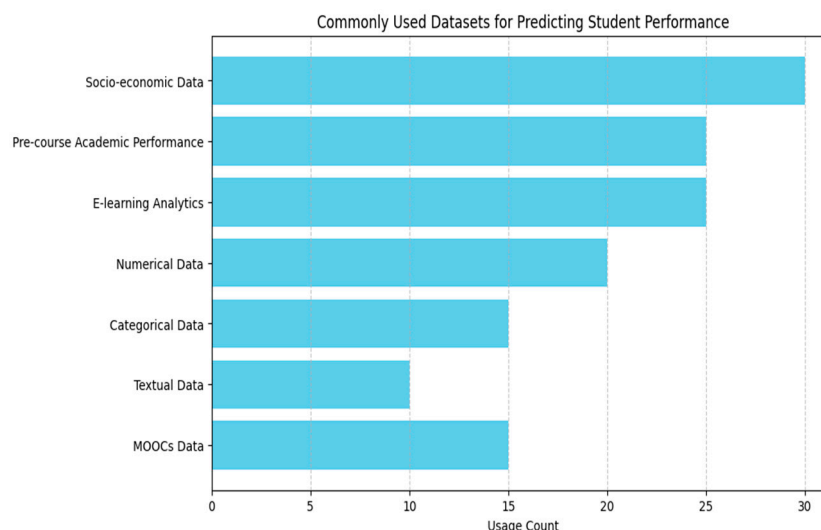


Figure 2. Commonly used datasets for predicting student performance.

The use of socio-economic and academic features is consistent with prior reviews, which emphasise the importance of holistic student information for accurate predictions [9,78]. Additionally, the growing reliance on LMS data, particularly in the context of MOOCs, aligns with recent trends that highlight the value of tracking student interactions and behaviour in online learning environments [79].

The review also corroborates earlier findings that predictive models increasingly integrate behavioural data, such as study habits and class participation, which provide further insights into student engagement and potential performance risks [72,80]. However, the analysis also reveals that textual data from student comments, although used in a minority of studies, are an emerging area that warrants further exploration.

While this review highlights commonly used datasets and their characteristics, such as socio-economic, academic, and behavioural metrics, the importance of evaluating the volumes of datasets used in predictive models for education should not be overlooked. Unfortunately, many of the included studies did not provide specific details on the dataset volumes or their diversity, making it challenging to assess their sufficiency comprehensively. However, the review does emphasise the need for diverse and representative datasets to improve the generalisability of predictive models. Future research should focus on systematically evaluating dataset sizes and their demographic coverage to ensure that models are robust, scalable, and applicable across various educational settings.

4.3. Key Factors Influencing the Accuracy of Educational Predictive Models

This section discusses the key factors that influence the accuracy of predictive models in an educational context, based on the findings of this systematic review. The analysis identifies several critical elements, including algorithm selection, data preprocessing, feature selection, and the inclusion of demographic and behavioural factors. These findings provide insights into how predictive models can be enhanced for better accuracy in educational settings.

A range of factors significantly influence the precision of educational predictive models. The choice of machine learning algorithms, such as Gradient Boosting, Random Forest, and neural networks, plays a crucial role. These algorithms are particularly effective in handling non-linear relationships and high-dimensional data, which are common in educational datasets [19]. Moreover, the quality and completeness of the data, including appropriate preprocessing techniques, such as normalisation and imputation, are essential for ensuring model reliability [19,20].

Feature selection methods, such as Principal Component Analysis (PCA) and correlation-based approaches, enhance the predictive power by focusing on the most influential factors, such as demographic, behavioural, and academic metrics [19,21]. Cross-validation techniques and sophisticated data handling methods, such as the SMOTE filter, further improve model performance, particularly in dealing with unbalanced datasets [21].

The algorithms shown in Figure 3 reflect the diversity of the methods employed in educational predictive modelling, including the most commonly used algorithms such as SVMs, ANNs, and Decision Trees. These algorithms are essential in addressing the complexities inherent in educational data, thereby improving the predictive accuracy of the models.

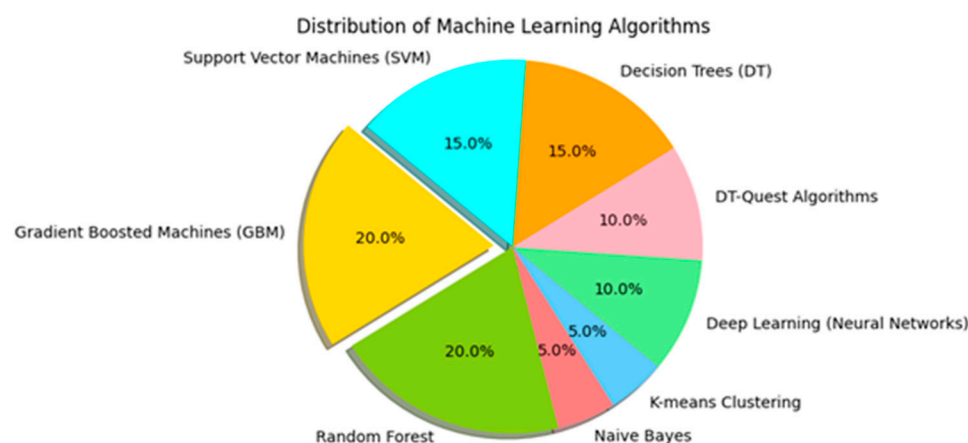


Figure 3. The most used algorithms in educational predictive modelling.

The predictive accuracy of models is also influenced by demographic and behavioural factors. Demographic variables such as race, gender, socio-economic status, and native language provide important context for understanding student performance, though they also raise concerns about potential biases [81]. Other factors, such as age, geographic location, and academic qualifications, further shape student engagement and access to educational resources, impacting the accuracy of predictions [82,83].

Behavioural factors, including class attendance, study habits, interaction with LMS, and participation in class activities, offer real-time indicators of student engagement. These behaviours significantly enhance the ability of the model to predict academic outcomes and identify at-risk students [84,85]. Combining demographic and behavioural data results in more robust models capable of addressing the multifaceted nature of student success [82,85].

The completeness, quality, and granularity of the data are critical for the accuracy of predictive models. Missing or inaccurate data can distort model outcomes, making effective imputation techniques essential [55,86]. Data granularity, such as detailed logs of student interactions with LMS, enables models to capture nuanced patterns in student behaviour, further improving prediction accuracy [86,87].

Contextual relevance is also paramount, ensuring that data align with the specific instructional design and learning environment [87,88]. The variability in LMS usage across different courses highlights the importance of using contextually appropriate data to avoid misleading predictions [88].

The selection of algorithms plays a fundamental role in model accuracy. Random Forest and Gradient Boosted Machines (GBMs) are widely recognised for their ability to handle diverse data types and reduce overfitting through ensemble learning [58,89]. SVMs and Decision Trees are also effective, particularly in capturing complex patterns in high-dimensional data [89,90].

Deep learning models, especially neural networks, excel in modelling non-linear relationships but require substantial computational resources [58]. Ensemble methods, such as those combining NB, k-Nearest Neighbors (k-NN), and Decision Trees, offer further improvements by leveraging the strengths of multiple algorithms [91].

4.4. Role of ML Prediction Models in Enhancing Decision-Making in Education

This section explores how prediction models, through machine learning algorithms and data processing techniques, enable accurate decision-making in educational settings. The discussion highlights the role of advanced machine learning algorithms, the importance of data preprocessing, integration with analytical tools, and the application of these models across various domains.

Advanced ML algorithms are essential for improving decision-making in education. SVM, Decision Trees, ANN, and ensemble methods, such as Random Forest and Gradient Boosting, significantly enhance the accuracy of prediction models. These algorithms are particularly adept at managing complex and non-linear relationships in educational data, making them suitable for forecasting student performance and behaviour [22]. This capability allows educators to make informed decisions, such as identifying students at risk of academic failure and implementing early interventions to improve outcomes [92,93].

In addition, the use of automated machine learning (AutoML) systems simplifies the process of model selection and hyperparameter tuning, which further improves the precision of predictions [10]. Techniques such as cross-validation and sensitivity analysis ensure the continuous refinement and reliability of these models, supporting data-driven decision-making in educational contexts [71,94].

By leveraging these sophisticated algorithms, educational institutions can optimise resource allocation, plan strategically, and provide timely support to students. This ultimately leads to improved educational outcomes and a more effective use of institutional resources.

Accurate decision-making in educational contexts heavily relies on robust data processing techniques in machine learning models. Proper data preprocessing is critical to ensure that the models operate on high-quality, reliable data. This involves addressing issues such as missing or inaccurate data, which can skew predictions if left unchecked [54].

Normalisation—scaling all variables—improves model accuracy by ensuring that different types of data are comparable [95]. Effective data preprocessing also includes feature selection, which identifies the most relevant predictors, and cross-validation, which assesses the reliability and generalisability of models [96]. These techniques ensure that predictions are not overfitted to specific datasets, improving their application across diverse educational settings.

Through proper data processing, institutions can develop predictive models that reliably identify at-risk students, helping to improve retention rates and student success [96]. This also supports more informed decision-making in resource allocation and targeted intervention strategies.

Machine learning models, when integrated with advanced analytical tools, further enhance decision-making accuracy by providing a comprehensive approach to analysing vast educational datasets. Predictive models using techniques such as SVM, neural networks, and Decision Trees can detect students at risk of underperformance or dropout by analysing complex socio-demographic and academic data [97].

These tools enable institutions to uncover hidden patterns and correlations in the data, facilitating the early detection of at-risk students and the implementation of targeted interventions [98]. Moreover, integrating machine learning models with analytical platforms provides actionable insights, improving overall decision-making processes in

education [99]. By enhancing prediction models with these tools, institutions can optimise resource allocation and intervention strategies, resulting in improved student outcomes and retention rates.

The application of machine learning models in education, across various domains, significantly enhances decision-making by allowing institutions to process large, complex datasets with greater precision. Algorithms such as Random Forest Classifiers, Artificial Neural Networks, and SVMs enable the incorporation of diverse data sources—ranging from socio-economic variables to academic performance metrics—into a unified framework that accurately forecasts student outcomes [100].

By analysing such varied data, educational institutions can develop targeted policies and interventions, identify at-risk students earlier, and optimise resource allocation [101]. In addition, the cross-domain applicability of these models allows for the adaptation of best practices from fields, such as healthcare [102], finance [97], and cybersecurity [103], into educational contexts, further improving decision-making and institutional efficiency [104].

4.5. Managing Uncertainty and Variability in Educational Prediction Models

This section explores how prediction models in education address uncertainty and variability in their forecasts. Several techniques, including Bayesian approaches, probabilistic forecasting, and post hoc sampling strategies, are used to enhance model reliability and inform decision-making in complex educational environments.

Prediction models address uncertainty and variability using advanced techniques, such as Bayesian probabilistic forecasting and EBLP. Bayesian approaches are particularly effective as they incorporate prior distributions and model averaging, offering a range of possible outcomes rather than a single forecast. This method increases predictive accuracy by accounting for uncertainties in model parameters [23].

EBLP improves forecast reliability by combining data across grade levels, academic years, and courses, reducing mean squared error (MSE) and stabilising predictions in smaller schools with more variability [24]. In addition, techniques such as probabilistic forecasting and Monte Carlo Simulation provide confidence intervals and simulate multiple scenarios, offering educators insights for decision-making in dynamic educational contexts [105,106]. These approaches ensure robust, reliable predictions, even in the presence of uncertainty.

To effectively measure uncertainty, prediction models employ techniques such as Bayesian Model Averaging (BMA), which calculates probability distribution by combining multiple models. BMA provides a comprehensive view of potential outcomes by considering variance both within and across models [107]. This technique also helps to distinguish between expected uncertainty (anticipated variability such as student performance fluctuations) and unexpected uncertainty (unforeseen changes, such as curriculum shifts) [108].

By updating predictions dynamically with new data, Bayesian frameworks further refine accuracy over time. Techniques such as Quantile Regression address heteroscedasticity by quantifying variability across different outcome levels, ensuring that predictions are robust in diverse scenarios [107]. These strategies collectively allow educational institutions to make well-informed decisions by accounting for both expected and unexpected uncertainties.

Handling data uncertainty in educational models requires advanced strategies to manage the inherent variability in educational data. AI, combined with big data analytics, enables models to process complex datasets and adapt to changing environments, which is crucial in contexts where student demographics and learning environments can introduce significant variability [109].

In addition, fuzzy logic is used to handle imprecise or ambiguous data by assigning degrees of truth rather than binary outcomes. This method is particularly effective in addressing the complexities of educational data, which often cannot be clearly classified [110]. Ensemble methods, which aggregate results from several models, reduce uncertainty by

smoothing individual model errors and leading to more robust predictions [100]. These approaches strengthen the ability of the model to handle data uncertainty, resulting in more accurate and practical recommendations for decision-making.

Post hoc sampling strategies play a crucial role in identifying biases and uncertainties in educational prediction models. By using Bayesian factors to categorise data, researchers can differentiate between true effects and random variations, thereby mitigating the risk of misleading conclusions caused by phenomena such as regression to the mean [111].

Cross-validation techniques such as k-fold cross-validation assess the performance of the model across different data subsets, revealing any biases that may arise in specific circumstances [112]. Sensitivity analysis evaluates how changes in input data influence model outcomes, highlighting key factors contributing to uncertainty [113]. Together, these strategies refine the accuracy of the model by addressing potential biases and enhancing its reliability in educational applications.

4.6. Comparing Machine Learning Algorithms to Traditional Statistical Models in Predicting Student Outcomes

This section discusses the comparative advantages of ML algorithms over traditional statistical models in predicting student outcomes based on the findings from various studies (see Section 3.3). The discussion focuses on the superiority of ML algorithms, their effectiveness in prediction, and their precision and reliability in handling educational data.

Machine learning algorithms consistently outperform traditional statistical models in predicting student outcomes, particularly due to their ability to manage large, complex datasets and capture non-linear relationships within the data. Traditional statistical models, such as linear regression, often rely on predefined assumptions about data distributions, limiting their flexibility in handling the diverse and multidimensional data typical in educational contexts [25].

In contrast, ML algorithms, including ANNs, Decision Trees, and SVMs, dynamically adjust their predictions based on the data, improving accuracy as new patterns emerge [25]. The adaptability of ML models allows them to accommodate a broader range of variables, making them particularly effective in settings such as massive open online courses (MOOCs) and hybrid learning environments. Advanced models, such as CNNs and LSTMs, further enhance the ability of ML algorithms to detect complex patterns in student behaviour. These capabilities make ML algorithms more precise and reliable compared to traditional models when forecasting student outcomes.

Machine learning algorithms are increasingly favoured over traditional statistical methods due to their superior ability to handle intricate, non-linear relationships in educational data. Conventional models, such as Decision Trees and logistic regression, often struggle to account for the complex interactions between diverse factors, including prior academic records, learning behaviours, and socio-demographic variables [114,115].

In contrast, ML techniques, including deep learning and ensemble methods, excel at processing vast datasets and uncovering hidden patterns that traditional models may overlook [116,117]. Studies show that ML algorithms achieve notably higher prediction accuracy across diverse educational settings, including online courses and traditional classroom environments [59,118].

The flexibility of ML models allows them to adapt to new data and improve their performance over time, further enhancing their predictive capabilities [119]. This dynamic approach not only leads to more accurate predictions but also facilitates more informed decision-making in education, emphasising the growing importance of ML in analysing student outcomes.

ML algorithms have consistently demonstrated greater precision and reliability than traditional statistical models in predicting student outcomes, especially in dealing with large and complex datasets. For instance, ANNs have been shown to outperform traditional methods, such as Decision Trees and logistic regression, when identifying patterns in student performance [119].

ML models such as Random Forest and SVMs, especially when used with advanced resampling techniques such as SVM-SMOTE, demonstrate significant efficacy in addressing the challenge of imbalanced datasets, a pervasive issue in educational data [57]. Imbalanced datasets occur when one class (e.g., at-risk students) is underrepresented compared to others (e.g., students performing well), which can lead to biased models that fail to accurately identify minority class instances. This challenge is critical in education, where accurately identifying at-risk students is necessary for timely interventions and support. To address this issue, techniques such as SMOTE (Synthetic Minority Over-sampling Technique) generate synthetic data points for the minority class by interpolating between existing samples, thereby balancing the dataset without duplicating instances. Variants such as SVM-SMOTE integrate SVMs to create synthetic samples closer to the decision boundary, improving classification performance in more complex data distributions. These techniques enhance the validity and generalisability of models by ensuring that minority classes are well represented during training, reducing the risk of overfitting and improving the ability of the model to generalise to new data.

In addition to SMOTE, other approaches, such as ADASYN (Adaptive Synthetic Sampling), focus on generating more synthetic samples in regions with higher difficulty, further improving the sensitivity of the model to nuanced patterns in minority class data. Cost-sensitive learning, another effective technique, penalises misclassification of the minority class more heavily, pushing the model to learn more robust decision boundaries. By incorporating these balancing strategies, models such as Random Forest and SVMs can deliver more reliable and equitable predictions, ultimately enhancing their application in educational contexts where detecting and supporting at-risk students is of paramount importance.

DNNs have also been found to offer superior accuracy and recall, particularly in early detection of at-risk students, allowing for timely interventions [120]. Additionally, models such as Naïve Bayes and ensemble methods such as StackingC have proven to be highly accurate, even with limited input data, further highlighting the reliability of ML algorithms in educational contexts where early identification of performance issues is critical [121].

4.7. Risks and Challenges of Using Prediction Models in Education

While prediction models offer significant potential to improve educational outcomes, they also present several risks and challenges that must be carefully managed to avoid unintended consequences. This discussion outlines key concerns related to bias, privacy, transparency, misclassification, and ethical considerations.

One of the most prominent risks in using prediction models, particularly those powered by AI, is the potential for perpetuating and amplifying existing biases. Since many AI systems learn from historical data, they can unintentionally encode patterns of inequality related to race, gender, and socio-economic status, reinforcing educational disparities [26]. For instance, if historical data reflect unequal treatment of certain student groups, the models may replicate these biases in future predictions, unfairly disadvantaging marginalised students. This presents a significant ethical challenge, requiring rigorous testing and bias mitigation strategies to ensure fairness in predictive outcomes.

Another critical challenge in implementing prediction models in education is the need for large volumes of personal data, including learning behaviours and sensitive demographic information. If not properly safeguarded, these data could be exposed to security breaches, leading to privacy violations and the misuse of student information [27]. Such breaches not only undermine trust in the educational institution but also raise ethical concerns regarding student consent and data protection. Institutions must implement strict data governance frameworks and comply with relevant privacy regulations to protect student data from misuse.

The complexity of AI-driven prediction models often results in “black box” systems, where the decision-making process is opaque and difficult for educators, students, and administrators to understand. This lack of transparency can lead to distrust in the recommendations of the model and reluctance to adopt the technology [122]. In addition,

when human judgment is sidelined in favour of algorithmic decisions, it can create a sense of loss of control and accountability in educational institutions. Efforts to improve the interpretability and explainability of AI models are crucial for building trust and ensuring that these tools are accepted and integrated into educational practices.

Prediction models, especially those working with incomplete, noisy, or biased data, are susceptible to misclassification. Incorrect predictions can result in inappropriate interventions that may stigmatise students or overlook those who genuinely need support. For example, students misclassified as at-risk may receive unnecessary interventions, while others may be neglected, exacerbating existing inequalities [123]. This risk is especially concerning when predictive models are used in high-stakes decision-making, where the consequences of errors are severe. Continuous monitoring and validation of prediction models are necessary to minimise these risks and improve reliability.

Prediction models in education also raise important ethical questions, particularly regarding the emphasis on quantifiable outcomes. Overreliance on data-driven predictions risks neglecting the qualitative aspects of education that contribute to holistic student development, such as creativity, emotional well-being, and interpersonal skills [124]. Ethical frameworks must be developed to ensure that these models do not prioritise short-term academic performance at the expense of broader educational goals. In addition, the role of human judgment in educational decision-making should be preserved to ensure that these models support, rather than replace, educators' professional expertise.

5. Limitations and Implications for Future Research

While prediction models in education show significant promise, there are notable limitations that need to be addressed for their effective and equitable implementation. One of the primary limitations is the reliance on historical data, which may encode and perpetuate existing biases related to race, gender, and socio-economic status. This can result in unfair predictions, exacerbating disparities rather than alleviating them. In addition, many of the reviewed studies rely on institution-specific datasets, limiting the generalisability of the findings to diverse educational contexts. The variability in data preprocessing methods, such as normalisation and feature selection, further complicates comparisons between models, making it difficult to assess overall performance across different settings.

Privacy concerns are another critical challenge, as the vast amount of personal data required by these models raises risks of data breaches and misuse. Moreover, many machine learning algorithms operate as "black box" systems, lacking transparency in their decision-making processes. This lack of interpretability can hinder trust in the predictions of the models, especially when human judgment is sidelined in favour of algorithmic decisions.

Misclassification is also a key risk, especially when models rely on incomplete or biased data. Incorrect predictions can lead to inappropriate interventions, stigmatising students or neglecting those who truly need support. Ethical considerations must be carefully addressed, particularly the risk of over-emphasising quantifiable outcomes at the expense of broader educational goals such as creativity, emotional well-being, and holistic development.

Given these limitations, future research should focus on several key areas to improve the effectiveness and fairness of prediction models. First, efforts should be made to standardise data collection and preprocessing methods to ensure more consistent and comparable results across different educational settings. In addition, there is a need for more inclusive datasets that reflect diverse student populations to enhance the generalisability of prediction models.

Bias detection and mitigation techniques require further exploration to ensure that these models do not perpetuate systemic inequalities. Future studies should investigate how cognitive, motivational, and non-cognitive factors can be integrated into prediction models to provide a more holistic understanding of student success.

Moreover, research should focus on developing more transparent and interpretable models to increase trust and acceptance among educators and policymakers. This could include the refinement of fairness-aware algorithms and the development of techniques that make machine learning models more explainable. Finally, ethical considerations surrounding data privacy and the potential misuse of predictive data should be a central focus, with the goal of creating models that are not only accurate but also responsible and equitable.

By addressing these limitations and pursuing these areas of research, prediction models in education can become more robust, transparent, and fair, ultimately leading to more effective interventions and improved student outcomes.

6. Conclusions

Following a comprehensive systematic review of prediction models in education, several key insights have emerged. This review demonstrates that ML algorithms, including SVMs, ANNs, and Decision Trees, consistently outperform traditional statistical models. These ML models excel in handling complex, non-linear educational data, adapting to various educational settings, and continuously improving their predictive accuracy. Their capability to analyse large, diverse datasets enables data-driven decisions that optimise educational outcomes.

The application of ML algorithms in education facilitates more accurate data-driven decision-making by incorporating socio-economic, demographic, and academic variables. Techniques such as Bayesian probabilistic forecasting further enhance the robustness of predictions. However, this review highlights that while these models offer improved precision, challenges related to bias, transparency, and ethical data handling remain significant.

Despite the potential of ML-based models, several limitations persist. The reliance on historical data risks perpetuating biases related to race, gender, and socio-economic status. The complexity of certain ML models also limits their interpretability, which can hinder their widespread adoption in educational institutions. Moreover, the quality of data inputs and variations in data processing methods across studies reduce the generalisability of current models.

Future research should prioritise the development of more transparent and interpretable models, ensuring that they are free from biases and align with ethical standards. Standardising data collection methods and incorporating non-traditional data, such as motivational and behavioural factors, could further improve the accuracy and fairness of prediction models. In addition, efforts to enhance the transparency of AI-driven models will be key to fostering trust among educators and stakeholders.

In conclusion, this review underscores the transformative potential of machine learning-based prediction models in education. By leveraging advanced algorithms and diverse datasets, these models provide valuable insights for improving student outcomes and institutional decision-making. However, to fully realise this potential, addressing ethical concerns, ensuring transparency, and mitigating biases will be essential. Future research and implementation efforts must focus on these aspects to ensure that educational prediction models are equitable, accurate, and effective across various educational contexts.

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Appendix A

Table A1. Summary of systematic review questions.

Systematic Review Questions	Question Themes	Number of Articles
What are the key reasons for using prediction models in education, and how do they impact student outcomes?	Purpose of Prediction Models in Education	5
	Impact on Student Outcomes	10
	Data-Driven Insights	4
	Proactive Student Support	4
What are the most used datasets for predicting student performance in educational predictive models?	Commonly Used Datasets	4
	Sources of Data	4
	Features and Characteristics of Datasets	6
	Techniques for Enhancing Predictive Models	3
What key factors influence the accuracy of predictive models in an educational context?	Key factors influence the accuracy of predictive models	3
	Demographic and Behavioural Factors	5
	Data Characteristics	4
	Algorithm Selection	9
	Specific Attributes	6
	Model Development	7
How do prediction models in machine learning enable accurate decision-making in education?	Role of Advanced ML Algorithms	6
	Data Processing Techniques	5
	Integration with Analytical Tools	4
	Application Across Domains	8
How do prediction models account for uncertainty and variability in their forecasts in education?	Estimating Predictive Uncertainty	4
	Measuring Uncertainty	3
	Handling Data Uncertainty	3
	Post Hoc Sampling Strategies	3
How do machine learning algorithms compare to traditional statistical models in predicting student outcomes?	Superiority of ML Algorithms	1
	Effectiveness in Prediction	7
	Precision and Reliability	4
What are the possible risks and challenges of using prediction models in education?	Risks	4
	Challenges	1

Table A2. Distribution of included studies across databases and corresponding references.

ScienceDirect (n = 43)	IEEE Xplore (n = 16)	ACM Digital Library (n = 2)	Google Scholar via ACM (n = 63)
[10,16,23,25,27,29,35,36,42,48,49,51,57,64–66,68,70,76,78,82,85–87,90,95,97,99–101,103,105,107,108,110,111,113–116,123].	[1,4,9,19,20,22,34,44,47,56,67,76,93,109,112,119]	[61,124]	[2,3,5–8,11–15,17,18,21,24,26,28,30–33,37–41,43,45,46,50,52–55,58,60,63,69,71–74,77,79–81,83,84,88,89,91,92,94,96,98,102,104,106,117,118,120–122]

Table A3. Study characteristics.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
1	[28]	2014	Quantitative study	K-12 and undergraduate students	Academic performance datasets	Machine Learning Techniques	Machine learning techniques enable early prediction of academic performance, supporting timely interventions for at-risk students.
2	[29]	2018	Quantitative study	Students in higher education	Institutional data	Heterogeneous Ensemble Models	Multi-model ensemble improves prediction accuracy for academic performance.
3	[26]	2022	Literature review	K-12 education stakeholders	Ethical challenges in AI integration	N/A	The study addresses ethical challenges associated with the adoption of artificial intelligence in K-12 education, focusing on fairness, transparency, and accountability in AI tools.
4	[53]	2021	Case study	K-12 students	Academic and attendance data	Neural Networks	Attendance records combined with academic data yield robust predictions in early education.
5	[9]	2021	Systematic review	Higher education institutions	LMS data, UC Irvine datasets	Explainable ML Models	Explainable ML improves transparency and decision-making in education.
6	[58]	2022	Case study	Online learners	MOOC engagement data	Gradient Boosting	Gradient Boosting models excel in predicting student retention in MOOCs.
7	[30]	2014	Quantitative study	First-year STEM students	Academic performance and demographic data	Hybrid Machine Learning Model	The hybrid model accurately predicts first-year retention in STEM disciplines, demonstrating the effectiveness of integrating multiple machine learning techniques for retention analysis.
8	[31]	2019	Quantitative study	University students	Academic performance datasets	Machine Learning Techniques	Machine learning techniques provide robust predictions for student performance, enabling data-driven educational interventions.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
9	[89]	2020	Literature review	Higher education students	Academic and demographic datasets	Various predictive models	The review identifies best practices for predicting academic success in higher education, highlighting the importance of data quality, model selection, and contextual factors in achieving high predictive accuracy.
10	[83]	2023	Literature review	STEM students	Interactive platform data	Logistic Regression	Logistic regression identifies key predictors of student engagement.
11	[88]	2023	Quantitative study	STEM students	Lab performance data	Gradient Boosting	Lab performance is a strong predictor of academic outcomes in STEM courses.
12	[27]	2021	Quantitative analysis	Students in e-learning platforms	Educational Big Data	Deep Learning	Integration of big data enables scalable and precise educational predictions.
13	[80]	2023	Literature review	Multi-institution cohort	Engagement data and completion rates	Ensemble Learning	Ensemble models drive actionable insights from large multi-institution datasets.
14	[5]	2021	Quantitative study	Technology management education learners	Innovation and technology datasets	Predictive Modelling	Predictive models provide actionable insights for identifying trends and opportunities in future technology development, emphasising the role of predictive analytics in technology management education.
15	[120]	2006	Quantitative study	Programming students	Academic performance and behavioural data	Statistical Models, Machine Learning Models	Both statistical and machine learning models demonstrate effective prediction of programming performance, with machine learning models providing greater flexibility for diverse datasets.
16	[66]	2022	Quantitative study	University students	Demographic and academic performance datasets	Statistical and Machine Learning Models	Demographic and academic features play a significant role in predicting student performance, supporting targeted interventions.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
17	[54]	2021	Quantitative study	Higher education students	Academic performance and behavioural datasets	Logistic Regression, Random Forest, Gradient Boosting, Neural Networks	A systematic comparison reveals trade-offs between interpretability and accuracy among predictive modelling methods, with Random Forest and Gradient Boosting achieving higher predictive power but logistic regression providing greater transparency.
18	[121]	2021	Literature review	AI ethics students	Case studies and educational programmes	Ethical Frameworks	Highlighted the importance of AI ethics education in addressing emerging challenges and ethical dilemmas in AI.
19	[32]	2019	Quantitative study	Higher education students	Academic performance datasets	Machine Learning Algorithms (e.g., Decision Trees, SVM, Random Forest, Neural Networks)	Comparative analysis highlights that Random Forest and neural networks achieve higher accuracy compared to other machine learning algorithms in predicting student performance.
20	[33]	2014	Quantitative study	Learners in higher education	Educational time-series data	Predictive Modelling	Developed a generalisable method for building predictive models of learner success using time-series educational data.
21	[98]	2019	Quantitative study	Computer engineering students	Academic performance data	Machine Learning Algorithms	Machine learning algorithms effectively predict academic performance for computer engineering students, emphasising the potential of AI in higher education analytics.
22	[94]	2020	Quantitative study	Higher education students	Retention and academic performance data	Data Mining, Machine Learning Models	Data mining and machine learning retention models provide actionable insights for improving student retention strategies, with specific algorithms demonstrating high predictive accuracy.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
23	[85]	2021	Quantitative study	Secondary school students	Online class attendance and performance data	Probabilistic Models	Predictive models accurately estimate the pass probability of secondary school students in online learning environments, supporting early interventions.
24	[8]	2024	Quantitative study	College English learners	College English Grade 4 results	Multi-model Prediction (e.g., Decision Trees, SVM, Neural Networks)	Multi-model prediction approaches improve the accuracy and robustness of grade forecasting for College English Grade 4, highlighting the advantages of combining multiple predictive techniques.
25	[74]	2014	Case study	Higher education students	Academic and behavioural datasets	Risk Assessment Predictive Models	The proactive risk assessment model offers promise in identifying at-risk students early but highlights challenges in balancing predictive accuracy with ethical considerations in higher education.
26	[34]	2020	Case study	University students	Demographic and socio-economic data	Decision Trees, Logistic Regression	Socio-economic factors have a significant impact on student performance predictions.
27	[71]	2018	Case study	Secondary and tertiary education students	Classroom clicker data	Logistic Regression	Low-cost predictive models using classroom clicker data effectively identify at-risk students, enabling proactive interventions.
28	[106]	2011	Quantitative study	K-12 students	Uncertainty measures	Bayesian Model Averaging	Bayesian methods effectively measure variability in academic performance predictions.
29	[117]	2018	Quantitative study	First-year engineering students	Academic and demographic datasets	Traditional Statistics, Machine Learning	Machine learning models outperform traditional statistical methods in predicting first-year engineering student success.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
30	[104]	2020	Quantitative study	Project management students and professionals	Project control datasets	Monte Carlo Simulation	Monte Carlo simulation effectively evaluates and compares forecasting model performance in project control settings.
31	[55]	2021	Quantitative study	College students in Saudi Arabia	Academic performance and demographic datasets	Data-Driven Predictive Models	Data-driven models accurately predict academic success, emphasising the impact of demographic and academic variables.
32	[14]	2021	Quantitative study	Researchers in Morocco	Research reform and performance datasets	Hybrid LSTM with Self-Attention Mechanism	Hybrid LSTM with self-attention mechanism effectively forecasts scientific research reform trends and outcomes.
33	[35]	2019	Quantitative study	Medical students	Academic performance and demographic datasets	Hybrid Data Mining Approach	A hybrid data mining approach effectively predicts academic performance, identifying key demographic and academic factors influencing outcomes.
34	[87]	2016	Case study	University students	Learning Analytics Data	Multiple Linear Regression	Customised analytics outperform one-size-fits-all prediction models.
35	[56]	2020	Case study	College students	Imbalanced academic datasets	SMOTE + Random Forest	SMOTE balancing improves predictive accuracy in handling imbalanced datasets.
36	[68]	2020	Quantitative study	First-year higher education students	Academic performance and demographic datasets	Gradient Boosting, Random Forest	Predictive models accurately identify first-year students at risk of academic failure, enabling targeted interventions.
37	[65]	2019	Quantitative study	Undergraduate students	Early engagement and academic performance data	Machine Learning	Early engagement metrics combined with machine learning models accurately predict student outcomes, enabling timely interventions.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
38	[72]	2021	Quantitative study	University students	Academic performance and engagement data	Machine Learning	Predictive models support proactive student interventions by accurately forecasting academic outcomes.
39	[21]	2022	Quantitative study	University students	Academic datasets with imbalanced classes	Machine Learning with Feature Selection	Feature selection approaches enhance prediction accuracy in machine learning models applied to imbalanced academic datasets.
40	[64]	2016	Quantitative study	University students	Institutional academic records	Decision Tree	Decision Trees offer interpretable and accurate predictions for academic outcomes.
41	[22]	2019	Quantitative study	K-12 students	Academic and engagement datasets	Machine Learning	Machine learning models provide educators with practical tools for predicting student performance in K-12 education, supporting timely interventions.
42	[90]	2019	Case study	Higher education teachers and students	Large-scale learning analytics data	Predictive Learning Analytics	Large-scale predictive learning analytics implementation highlights the critical role of teachers in interpreting and acting on predictions.
43	[99]	2021	Literature review	Educational researchers and practitioners	Diverse educational datasets	Machine Learning	Machine learning applications in education improve predictive accuracy, enabling data-driven decision-making and personalised interventions.
44	[57]	2022	Case study	MOOC participants	Interaction data from MOOCs	Gradient Boosting	Gradient Boosting identifies predictors of retention in MOOCs effectively.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
45	[15]	2019	Quantitative study	Higher education students	Behavioural data on procrastination	Educational Data Mining Models	Procrastination behaviour is a significant predictor of academic performance, with educational data mining models effectively identifying at-risk students based on behavioural patterns.
46	[36]	2013	Quantitative study	Engineering students	Course grades and engagement metrics	Logistic Regression, Decision Trees, Neural Networks, Bayesian Networks	Comparing models revealed that neural networks and Bayesian Networks performed better in predicting academic success in an engineering dynamics course.
47	[113]	2018	Quantitative study	Online learners	Learning session data	Machine Learning	Machine learning models effectively identify student difficulties during learning sessions, enabling targeted support and interventions.
48	[62]	2022	Quantitative study	Online engineering students	Academic performance and engagement datasets	Artificial Intelligence-Enabled Prediction Model	AI-enabled models accurately predict academic performance in online engineering education, supporting data-driven personalised interventions.
49	[86]	2021	Quantitative study	University students	Academic and instructional conditions data	Learning Analytics Models	Internal and instructional conditions significantly influence academic success predictions, emphasising the importance of student-centric learning analytics.
50	[79]	2017	Literature review	Students at risk	Academic performance datasets	Data Mining Techniques	Data mining techniques provide effective methods for predicting the academic performance of at-risk students, enabling timely interventions.
51	[23]	2021	Case study	K-12 students	Large-scale educational trend data (NAEP)	Bayesian Probabilistic Forecasting	Bayesian probabilistic models effectively forecast educational trends, providing insights for long-term planning in K-12 education.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
52	[59]	2015	Quantitative study	K-12 students	Academic performance and demographic datasets	Classification and Prediction Algorithms	Data mining algorithms accurately classify and predict slow learners, enabling targeted interventions in the education sector.
53	[37]	2021	Conceptual framework	Higher education students	Academic performance datasets	Machine Learning with Attribute Selection	Proposed a conceptual framework to guide attribute selection, improving the accuracy and interpretability of machine learning models for student performance prediction.
54	[38]	2024	Case study	MOOC participants	Interaction and completion rates	Gradient Boosting	Gradient Boosting identifies predictors of student success in MOOCs.
55	[39]	2023	Empirical study	Students across various disciplines	Academic performance and socio-economic data	Statistical Analysis, Regression Models	The study identifies key factors influencing academic performance, such as socio-economic status, attendance, and engagement, providing actionable insights for targeted interventions.
56	[16]	2009	Systematic literature review	Software engineering researchers	Systematic literature review methodology	N/A	Provides a structured approach for conducting systematic literature reviews in software engineering, enhancing evidence synthesis and reliability.
57	[60]	2017	Experimental	Undergraduate engineering students	Exam scores, attendance records	SVM	Early intervention based on poor attendance improves exam performance.
58	[110]	2021	Literature review	Higher education institutions	Academic and interaction datasets	Explainable AI	Transparent models enhance trust in AI-driven predictions.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
59	[4]	2023	Quantitative study	Autonomous robot systems researchers	Time-series and behavioural datasets	A hybrid approach combining Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) Neural Network	The hybrid CNN-LSTM model effectively predicts complex patterns in time-series data, showcasing improved accuracy for predictive tasks in AI and robotics education contexts.
60	[40]	2019	Experimental	College students	Learning management systems	Logistic Regression	LMS usage metrics strongly correlate with student success and retention.
61	[24]	2022	Bayesian probabilistic study	Students across grade levels	Aggregate student growth datasets	Empirical Best Linear Prediction (EBLP)	EBLP enhances accuracy and reduces variability in growth measures predictions.
62	[41].	2021	Quantitative study	K-12 students	Interaction metrics	Gradient Boosting	Predictive analytics reveals interaction patterns critical to academic success in early education.
63	[19]	2024	Quantitative study	Mixed educational settings	Institutional datasets	Gradient Boosting, Random Forest	Advanced ML models effectively handle high-dimensional data for performance predictions.
64	[96]	2023	Systematic review	Educational stakeholders	Various educational datasets	Artificial Intelligence Models	The systematic review highlights the applications of AI in educational analytics and decision-making, emphasising the role of AI models in enhancing predictive accuracy and supporting evidence-based strategies.
65	[42]	2021	Case study	High school students	Intelligent M-learning environments	Decision Trees	Predictive models facilitate academic success in mobile learning contexts.
66	[43]	2023	Quantitative study	K-12 students	Socio-economic and academic datasets	Decision Trees, Naïve Bayes	Socio-economic factors contribute significantly to early dropout predictions.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
67	[69]	2015	Quantitative study	Higher education students	Academic performance and intervention-level data	Data-Driven Predictive Models	The study demonstrates the effectiveness of data-driven predictive models in identifying intervention points to improve academic performance, enabling targeted strategies for student success.
68	[1]	2024	Experimental	Digital pathology practitioners	Whole-slide images (WSIs) and cloud-based data	Automated AI Models	ComPREPS, a cloud-based automated image analysis tool, democratizes AI usage in digital pathology, enhancing accessibility and precision in diagnostics.
69	[44]	2024	Quantitative study	University students	Academic performance and engagement data	Data Clustering and Predictive Modelling	Integrating data clustering with predictive modelling effectively monitors and forecasts student performance, aiding in timely educational interventions.
70	[81]	2022	Quantitative study	STEM students	Project-based datasets	Gradient Boosting	Project data provide robust insights into STEM academic outcomes.
71	[118]	2020	Quantitative study	Students across educational levels	Academic and behavioural datasets	Machine Learning Predictive Modelling	Machine learning models reliably identify key predictors of educational outcomes, supporting data-driven interventions and policy making.
72	[119]	2021	Quantitative study	Higher education students	Academic performance data (course grades)	Deep Neural Networks	Deep neural networks effectively predict students' academic performance based on their course grades, demonstrating high accuracy and scalability across diverse datasets.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
73	[45]	2018	Quantitative study	Preschool students	Temporal patterns of development test data	Predictive Modelling	Predictive models leveraging temporal patterns in development tests effectively forecast learning continuation in preschool education, providing actionable insights for early interventions.
74	[114]	2020	Systematic literature review	Higher education students	Learning analytics and academic performance data	Decision Trees, Neural Networks, SVM, Ensemble Models	Systematic review highlights the effectiveness of data mining and learning analytics techniques in predicting student performance, with ensemble models providing the highest accuracy.
75	[93]	2021	Quantitative study	Higher education institutions	Academic and decision-making datasets	Machine Learning	Ensuring correctness in machine learning-based decision-making enhances reliability and trustworthiness in higher educational institutions.
76	[91]	2020	Literature review	Students across educational levels	Academic performance datasets	Machine Learning Algorithms	Machine learning algorithms are effective in predicting student performance and improving learning outcomes, supporting data-driven educational strategies.
77	[18]	2021	Quantitative study	Students in higher education	Academic performance datasets	Machine Learning	Machine learning algorithms predict student academic performance with high accuracy, supporting proactive interventions in educational settings.
78	[67]	2020	Quantitative study	University students	Academic performance and risk-level datasets	Machine Learning	A predictive model using machine learning effectively forecasts student performance and identifies risk levels, supporting proactive interventions.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
79	[103]	2022	Quantitative study	STEM students	Academic performance and survey data	Ensemble Learning	Ensemble methods accurately predict factors influencing STEM students' intentions to enroll in chemistry-related courses, highlighting key engagement and demographic factors.
80	[92]	2022	Literature review	Students and educators across contexts	Academic forecasting datasets	Machine Learning	Highlights the prospects of machine learning for academic forecasting while identifying challenges such as data quality, model interpretability, and scalability.
81	[3]	2011	Quantitative study	Oil spill detection researchers	Remote sensing data	Artificial Bee Colony Algorithm, Artificial Neural Network	The Artificial Bee Colony algorithm effectively optimises the training of Artificial Neural Networks, enhancing accuracy and efficiency in oil spill detection.
82	[95]	2021	Quantitative study	Secondary school students	Interactive learning environments	Neural Networks	Neural networks effectively capture trends in interactive learning environments.
83	[84]	2020	Quantitative study	Learners in various educational settings	Demographic and behavioural datasets	Educational Data Mining (EDM) Models	Incorporating demographic data into EDM research improves the understanding of learning behaviours and outcomes, highlighting disparities and providing insights for equitable educational practices.
84	[100]	2023	Quantitative study	Higher education students	Cognitive, affective, and psychomotor learning data	Neuro-Fuzzy Models, Machine Learning Algorithms	Neuro-fuzzy models combined with machine learning algorithms accurately predict academic performance by integrating cognitive, affective, and psychomotor learning data.

Table A3. Cont.							
No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
85	[105]	2021	Quantitative study	School choice participants	Structural demand and school choice data	Structural Demand Models	Structural demand models provide accurate counterfactual predictions in school choice scenarios, supporting better policy making in educational planning.
86	[46]	2011	Quantitative study	Decision-making participants	Behavioural and decision-making datasets	Bayesian Learning	Bayesian learning effectively models decision-making in unstable settings, accounting for risk, unexpected uncertainty, and estimation uncertainty.
87	[6]	2020	Literature review	Clinical and educational model developers	Prediction model datasets	Prediction Models	Provides a comprehensive framework for the development, evaluation, and application of prediction models in clinical and educational contexts.
88	[47]	2021	Quantitative study	Programming students	Behavioural and performance data	Interpretable Black-Box Predictive Model	By interpreting black-box models, the study reveals insights into individual and collective programming students' behaviour, providing actionable explanations for improved teaching strategies.
89	[111]	2020	Quantitative study	Multi-institution cohort	Brain-behaviour interaction data	Neural Networks	Advanced Neural Networks predict behavioural outcomes from brain data.
90	[115]	2019	Quantitative study	Baccalaureate students in Morocco	Academic records and socio-demographic data	Machine Learning Framework	The proposed machine learning framework effectively predicts baccalaureate student performance, demonstrating the impact of socio-demographic factors on outcomes.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
91	[78]	2019	Quantitative study	MOOC participants	MOOC interaction and knowledge mastery datasets	Multi-Task Learning	Multi-task learning models effectively predict student performance and identify deficiencies in mastering knowledge points, supporting personalised learning in MOOCs.
92	[77]	2023	Quantitative study	University students	Academic performance datasets	Machine Learning	Machine learning techniques effectively predict students' final grades, enabling data-driven educational interventions and support strategies.
93	[17]	2010	Quantitative study	Educational data mining participants	Academic performance datasets	CHAID (Chi-squared Automatic Interaction Detector)	CHAID-based models provide interpretable and accurate predictions of student performance, enabling targeted interventions in education.
94	[48]	2022	Quantitative study	Students in various educational settings	Educational data mining datasets	Generic Predictive Models	Developing generic models for predicting student outcomes improves scalability and applicability across diverse educational datasets.
95	[7]	2016	Literature review	Clinical researchers	Clinical datasets	Clinical Prediction Models	The study reviews clinical prediction models, emphasising their development, validation, and application in healthcare settings to improve patient outcomes through data-driven decision-making.
96	[116]	2020	Literature review	Students across various educational levels	Academic performance datasets	Machine Learning	Machine learning methods are effective for analysing and predicting student performance, providing valuable insights for educational improvement.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
97	[97]	2022	Systematic review	Teacher education participants	Learning analytics and AI tools in teacher education	Artificial Intelligence, Learning Analytics Frameworks	The review highlights the integration of AI and learning analytics in teacher education, identifying key areas for improving teaching practices and student outcomes through predictive analytics.
98	[101]	2021	Literature review	Multi-domain data science practitioners	Data-driven decision-making datasets	Data Science and Analytics Frameworks	Provides an overview of data science and analytics frameworks, emphasising their role in smart computing, decision-making, and diverse applications, including education.
99	[25]	2021	Systematic literature review	Students across various levels	Academic performance prediction datasets	Machine Learning	Highlights critical gaps in machine learning applications for student performance prediction and suggests remedies for improving accuracy and generalisability.
100	[102]	2023	Quantitative study	University students	Educational data mining datasets	Machine Learning	The prediction model effectively uses educational data mining and machine learning techniques to forecast students' academic performance.
101	[49]	2012	Quantitative study	Secondary school students	Placement-test scores and demographic datasets	Data Mining	Data mining models effectively predict and analyse placement-test scores, identifying key factors influencing academic outcomes.
102	[20]	2023	Quantitative study	Learners in higher education	Multisource academic and behavioural data	Regression-Driven Predictive Model	A regression-driven predictive model effectively estimates learner performance by integrating multisource data, enabling accurate forecasting and interventions.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
103	[112]	2014	Quantitative study	Software defect prediction researchers	Software defect datasets	Machine Learning	Explores researcher bias in machine learning applications for software defect prediction and emphasises the importance of rigorous evaluation methodologies.
104	[50]	2024	Quantitative study	Higher education institutions	Academic performance and institutional data	Linear Regression, Decision Tree, Random Forest, Multilayer Perceptron (MLP)	The proposed framework integrates machine learning algorithms into performance management systems, with the Multilayer Perceptron (MLP) algorithm outperforming others (MSE: 0.018; MAE: 0.105; R ² : 0.842), enhancing organisational efficiency and student success.
105	[2]	2020	Quantitative study	Higher education students	Academic performance datasets	Hybrid Machine Learning Algorithms	Hybrid machine learning algorithms outperform traditional methods in predicting academic performance.
106	[107]	2019	Literature review	Neuroscience and behavioural study participants	Adaptive learning datasets	Adaptive Learning Models	Explores mechanisms of adaptive learning under expected and unexpected uncertainty, highlighting implications for decision-making and behaviour.
107	[122]	2018	Quantitative study	School administrators and students	Administrative and academic performance datasets	Big Data Predictive Models	Big data predictive models accurately identify students at risk of dropping out, enabling data-driven administrative interventions to improve retention.
108	[75]	2016	Quantitative study	Higher education students	Next-term performance datasets	Recommender System Approach	Recommender systems predict next-term student performance effectively, offering personalised recommendations for academic success.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
109	[82]	2014	Quantitative study	Mobile learning users	Behavioural intention data	Hybrid SEM–Neural Network Approach	The hybrid SEM–neural network approach accurately predicts behavioural intentions to use mobile learning, emphasising usability and perceived value.
110	[70]	2022	Quantitative study	College and university students	Academic performance and administrative datasets	Artificial Intelligence Models	Data-driven AI models enhance decision-making in higher education, improving efficiency and accuracy in administrative and academic processes.
111	[63]	2010	Experimental	University students	Academic performance datasets	Recommender System Models	Recommender system models are effective in predicting student performance and guiding personalised educational recommendations.
112	[76]	2009	Experimental	University students	Academic performance datasets with imbalanced classes	Machine Learning with Class Balancing Techniques	Addressing class imbalance significantly improves the accuracy of academic performance predictions, supporting better outcomes in educational data mining.
113	[108]	2020	Quantitative study	University students	Exam performance datasets	Hybrid Approach Combining Structural Equation Modelling (SEM) and ANNs	Comprehensive comparison shows neural networks and SVM achieve higher accuracy for predicting student exam performance compared to traditional models like Decision Trees.
114	[123]	2021	Systematic review	Higher education institutions	Predictive analytics datasets	Predictive Analytics Models	The review explores the opportunities, challenges, and future directions of predictive analytics in higher education, emphasising scalability and ethical considerations.
115	[73]	2019	Quantitative study	At-risk college students	Academic performance and faculty assessments	Predictive Model	Predictive models outperform faculty identification in detecting at-risk students, enabling effective advising strategies.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
116	[11]	2020	Quantitative study	VLE (Virtual Learning Environment) students	VLE big data	Deep Learning Models	Deep learning models effectively predict academic performance by analysing large-scale VLE data, outperforming traditional methods.
117	[109]	2016	Literature review	Big data analysts	Big data with uncertainty	Analytical Strategies	Discusses strategies for addressing uncertainty in big data analytics, highlighting future challenges and opportunities for enhanced decision-making.
118	[51]	2015	Predictive model development	University students (participation-based data)	Learning analytics and educational interaction data	Genetic Programming-based predictive models	Participation data predicts students' final performance effectively. Genetic Programming provides interpretable results integrating theory and data mining.
119	[52]	2022	Quantitative study	University students	Academic performance datasets	Machine Learning Algorithms	Machine learning algorithms predict academic outcomes with high accuracy, emphasising data-driven interventions.
120	[12]	2021	Quantitative study	Intelligent Tutoring System users	ITS interaction data	Machine Learning	Machine learning-based student modelling improves personalised learning in ITS environments.
121	[13]	2023	Experimental	College students	Feature extraction and interaction data	Attention-based BiGRU	Attention-based BiGRU enhances predictions of learning performance using interaction features.
122	[10]	2021	Quantitative study	University students	Academic performance datasets	Automated Machine Learning (AutoML)	AutoML approaches significantly enhance the prediction of student success, streamlining the process of model development and increasing accuracy.

Table A3. Cont.

No	Author(s)	Year	Design	Participants	Dataset(s)	Model(s)	Key Findings
123	[124]	2010	Case study	University students	Academic performance and retention data	Data Mining Techniques	Data mining methods effectively identify at-risk students and enable tailored interventions to improve retention in higher education institutions.
124	[61]	2008	Quantitative study	Students in remedial education	Web-based English learning data	Data Mining Models	Data mining models effectively predict performance in web-based remedial English education, providing insights into areas for improvement and personalised instruction.

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